

REPORT ON DATA CLEANING, PREPARATION.

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INTRODUCTION

This report presents the steps taken to clean and prepare a raw marketing dataset. The main aim is to improve the data quality by standardizing data formats, removing duplicate records, resolving missing data. These steps ensure that the dataset is consistent, free of errors, ready for deeper analysis and more accurate insight.

DATASET OBSERVATION

Before improving the data quality, the product dataset consists of 1200 rows and 14 columns which incorporating both categorical and numerical variables.

- The columns include;
 - OrderID, Dates, CustomerID, Product Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemInCart, CouponCode, ReferralSource, and TotalPrice
- Core issues observed before cleaning the dataset include;
 - Missing values in CouponCode Column, which could affect the accuracy of the further analysis.
 - Non-standardised or unclear column names.
 - Inconsistent Formatting were observed such as spacing and text casing.

BEFORE DATA CLEANING IMAGE

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
180	ORD20117	#####	C65107	Monitor	1	70.24	456	Main ! Credit Car	Cancelled	TRK102615	4	FREESHIP	Google	70.24						
181	ORD20117	#####	C84630	Laptop	5	518.55	907	Main ! Credit Car	Cancelled	TRK480545	6	WINTER15	Instagram	2592.75						
182	ORD20118	#####	C42235	Chair	4	298.36	722	Main ! Debit Car	Pending	TRK493791	8	FREESHIP	Google	1193.44						
183	ORD20118	#####	C70246	Desk	3	158.74	293	Main ! Online	Pending	TRK457081	7	FREESHIP	Instagram	476.22						
184	ORD20118	#####	C18473	Chair	2	420.29	700	Main ! Gift Card	Returned	TRK978342	3	SAVE10	Google	840.58						
185	ORD20118	#####	C12607	Desk	1	281.52	460	Main ! Gift Card	Shipped	TRK836531	6	WINTER15	Referral	281.52						
186	ORD20118	#####	C10222	Laptop	1	204.08	979	Main ! Gift Card	Returned	TRK862971	4	Google		204.08						
187	ORD20118	#####	C77344	Chair	1	343.31	777	Main ! Gift Card	Returned	TRK206054	6	SAVE10	Google	343.31						
188	ORD20118	#####	C65497	Monitor	1	163.32	676	Main ! Cash	Shipped	TRK837022	6	WINTER15	Facebook	163.32						
189	ORD20118	#####	C26340	Desk	2	184.6	562	Main ! Gift Card	Delivered	TRK599622	7	WINTER15	Email	369.2						
190	ORD20118	#####	C75619	Printer	2	371.29	924	Main ! Gift Card	Shipped	TRK230865	5	WINTER15	Facebook	742.58						
191	ORD20118	#####	C96136	Tablet	5	389.22	149	Main ! Cash	Cancelled	TRK608564	7	Referral		1946.1						
192	ORD20119	#####	C46886	Desk	3	616.39	754	Main ! Credit Car	Shipped	TRK827914	6	FREESHIP	Instagram	1849.17						
193	ORD20119	#####	C44148	Desk	4	376.43	809	Main ! Gift Card	Pending	TRK678722	5	WINTER15	Instagram	1505.72						
194	ORD20119	#####	C72007	Chair	3	432.85	999	Main ! Online	Pending	TRK162405	4	Instagram		1298.55						
195	ORD20119	#####	C45104	Tablet	1	687.89	348	Main ! Cash	Returned	TRK385591	1	FREESHIP	Email	687.89						
196	ORD20119	#####	C29885	Printer	3	189.44	725	Main ! Cash	Pending	TRK620671	4	FREESHIP	Email	568.32						
197	ORD20119	#####	C21126	Desk	1	107.04	392	Main ! Credit Car	Cancelled	TRK380092	6	FREESHIP	Google	107.04						
198	ORD20119	#####	C20095	Monitor	2	662.53	778	Main ! Online	Cancelled	TRK692075	5	Facebook		1325.06						
199	ORD20119	#####	C79674	Tablet	2	436.84	275	Main ! Online	Delivered	TRK880392	2	FREESHIP	Instagram	873.68						
200	ORD20119	#####	C64753	Chair	4	262.52	509	Main ! Debit Car	Cancelled	TRK716833	4	WINTER15	Instagram	1050.08						
201	ORD20119	#####	C57502	Tablet	4	560.58	201	Main ! Gift Card	Returned	TRK511162	6	SAVE10	Referral	2242.32						

DATA CLEANING STEPS

The dataset was imported into Power Query to support effective cleaning and further analysis. The steps applied were:

- Removing duplicate from the OrderID column to ensure data accuracy and reliability but no duplicate form, Rows stands as 1200 and 14 columns .
- I standardized all the Columns name by renaming them to ensure consistency (Capitalized Each Word) and spacing were used in this case. For example, 'ItemInCart' was renamed to 'Item In Cart'
- Missing Values in the Coupon Code column was replaced with "N/A" .
- The Date Column was Formatted to the right format from "MM/DD/YYYY" to "YYYY/MM/DD".

CLEANED DATASET IMAGE

Order ID	Date	Customer ID	Product	Quantity	Unit Price	Shipping Address	Payment Method	Order Status	Tracking Number	Items In Cart	Coupon Code	Referral Source
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	N/A	Email
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email
ORD200012	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral
ORD200013	2024-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	N/A	Google
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email
ORD200017	2024-03-02	C60672	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	N/A	Referral
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook
ORD200019	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram

CONCLUSION

The dataset is now clean, consistent, and ready for further marketing analysis. Key improvements included removing duplicate, handling missing values, date formatting, and standardizing columns while preserving key information in the Coupon Code Column.